Online Retail Sales Analysis Database

Database Tables:

Create table Customers (customer_id, name, city)

QUERY:

```
CREATE TABLE client_info (
   client_id INT PRIMARY KEY,
   client_name VARCHAR(50),
   location VARCHAR(50)
);
```

Output:

```
SQL> desc client_info
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 CLIENT_ID                                 NOT NULL NUMBER(38)
 CLIENT_NAME                                        VARCHAR2(50)
 LOCATION                                           VARCHAR2(50)
```

Create table Products (product_id, name, category, price)

Query:

```
CREATE TABLE item_catalog (
   item_id INT PRIMARY KEY,
   item_name VARCHAR(50),
   item_group VARCHAR(50),
   unit_price DECIMAL(10,2)
);
```

Output:

```
SQL> desc item_catalog
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 ITEM_ID                                   NOT NULL NUMBER(38)
 ITEM_NAME                                          VARCHAR2(50)
 ITEM_GROUP                                         VARCHAR2(50)
 UNIT_PRICE                                         NUMBER(10,2)

SQL> |
```

Create table Orders (order_id, customer_id, date)

Query:

```
CREATE TABLE order_log (
    order_id INT PRIMARY KEY,
    client_id INT,
    order_date DATE,
    FOREIGN KEY (client_id) REFERENCES client_info(client_id)
);
```

Output:

```
SQL> desc order_log
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 ORDER_ID                                  NOT NULL NUMBER(38)
 CLIENT_ID                                          NUMBER(38)
 ORDER_DATE                                         DATE

SQL> |
```

Create table Order_Items (order_id, product_id, quantity)

QUERY:

```
CREATE TABLE order_details (
    order_id INT,
    item_id INT,
    quantity INT,
    FOREIGN KEY (order_id) REFERENCES order_log(order_id),
    FOREIGN KEY (item_id) REFERENCES item_catalog(item_id)
);
```

Output:

```
SQL> desc order_details;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 ORDER_ID                                           NUMBER(38)
 ITEM_ID                                            NUMBER(38)
 QUANTITY                                           NUMBER(38)

SQL> |
```

Product_details:

Query:

INSERT ALL

INTO item_catalog VALUES (101, 'Wireless Mouse', 'Electronics', 599)

INTO item_catalog VALUES (102, 'Bluetooth Speaker', 'Electronics', 1499)

INTO item_catalog VALUES (103, 'Notebook Pack', 'Stationery', 299)

INTO item_catalog VALUES (104, 'Water Bottle', 'Accessories', 199)

INTO item_catalog VALUES (105, 'Backpack', 'Accessories', 999)

SELECT * FROM dual;

Output:

```
SQL> select * from item_catalog;

   ITEM_ID ITEM_NAME
---------- --------------------------------------------------
ITEM_GROUP                                         UNIT_PRICE
-------------------------------------------------- ----------
       101 Wireless Mouse
Electronics                                               599

       102 Bluetooth Speaker
Electronics                                              1499

       103 Notebook Pack
Stationery                                                299


   ITEM_ID ITEM_NAME
---------- --------------------------------------------------
ITEM_GROUP                                         UNIT_PRICE
-------------------------------------------------- ----------
       104 Water Bottle
Accessories                                               199

       105 Backpack
Accessories                                               999


SQL> |
```

Customer_details

Query: INSERT ALL

```
   INTO client_info VALUES (1, 'Rahul Sharma', 'Mumbai')
   INTO client_info VALUES (2, 'Sneha Reddy', 'Hyderabad')
   INTO client_info VALUES (3, 'Amit Verma', 'Delhi')
   INTO client_info VALUES (4, 'Priya Nair', 'Kochi')
   INTO client_info VALUES (5, 'Karan Mehta', 'Pune')
SELECT * FROM dual;
```

Output:

```
SQL> select * from client_info;

 CLIENT_ID CLIENT_NAME
---------- --------------------------------------------------
LOCATION
--------------------------------------------------
         1 Rahul Sharma
Mumbai

         2 Sneha Reddy
Hyderabad

         3 Amit Verma
Delhi


 CLIENT_ID CLIENT_NAME
---------- --------------------------------------------------
LOCATION
--------------------------------------------------
         4 Priya Nair
Kochi

         5 Karan Mehta
Pune
```

Orders:

Query:

```
INSERT ALL
   INTO order_details VALUES (201, 101, 2)
   INTO order_details VALUES (201, 103, 3)
   INTO order_details VALUES (202, 102, 1)
   INTO order_details VALUES (202, 104, 2)
   INTO order_details VALUES (203, 101, 1)
   INTO order_details VALUES (203, 105, 2)
```

INTO order_details VALUES (204, 102, 2)

INTO order_details VALUES (204, 103, 1)

INTO order_details VALUES (205, 104, 4)

SELECT * FROM dual;

Output:

```
SQL> select * from order_details;

  ORDER_ID    ITEM_ID   QUANTITY
---------- ---------- ----------
       201        101          2
       201        103          3
       202        102          1
       202        104          2
       203        101          1
       203        105          2
       204        102          2
       204        103          1
       205        104          4

9 rows selected.
```

Order_details:

Query:

INSERT ALL

INTO order_log VALUES (201, 1, DATE '2026-03-10')

INTO order_log VALUES (202, 2, DATE '2026-03-15')

INTO order_log VALUES (203, 3, DATE '2026-04-05')

INTO order_log VALUES (204, 1, DATE '2026-04-12')

INTO order_log VALUES (205, 4, DATE '2026-04-18')

SELECT * FROM dual;

Output:

```
SQL> select * from order_log;

  ORDER_ID  CLIENT_ID ORDER_DAT
---------- ---------- ---------
       201          1 10-MAR-26
       202          2 15-MAR-26
       203          3 05-APR-26
       204          1 12-APR-26
       205          4 18-APR-26

SQL> |
```

1.write a query to select Top-Selling Products:

Query:

SELECT i.item_name, SUM(d.quantity) AS total_sold

FROM order_details d

JOIN item_catalog i ON d.item_id = i.item_id

GROUP BY i.item_name

ORDER BY total_sold DESC;

Output:

```
SQL> SELECT i.item_name, SUM(d.quantity) AS total_sold
  2  FROM order_details d
  3  JOIN item_catalog i ON d.item_id = i.item_id
  4  GROUP BY i.item_name
  5  ORDER BY total_sold DESC;

ITEM_NAME                                          TOTAL_SOLD
-------------------------------------------------- ----------
Water Bottle                                                6
Notebook Pack                                               4
Wireless Mouse                                              3
Bluetooth Speaker                                           3
Backpack                                                    2
```

2:write a query to select most valuable customer

Query:

SELECT c.client_name,

SUM(d.quantity * i.unit_price) AS total_spent

FROM client_info c

JOIN order_log o ON c.client_id = o.client_id

JOIN order_details d ON o.order_id = d.order_id

JOIN item_catalog i ON d.item_id = i.item_id

GROUP BY c.client_name

ORDER BY total_spent DESC;

Output:

```
SQL> SELECT c.client_name,
  2         SUM(d.quantity * i.unit_price) AS total_spent
  3  FROM client_info c
  4  JOIN order_log o ON c.client_id = o.client_id
  5  JOIN order_details d ON o.order_id = d.order_id
  6  JOIN item_catalog i ON d.item_id = i.item_id
  7  GROUP BY c.client_name
  8  ORDER BY total_spent DESC;

CLIENT_NAME                                        TOTAL_SPENT
-------------------------------------------------- -----------
Rahul Sharma                                              5392
Amit Verma                                                2597
Sneha Reddy                                               1897
Priya Nair                                                 796
```

3:Write a query to select the Monthly Revenue

Query: SELECT DATE_FORMAT(o.order_date, '%Y-%m') AS month,

SUM(d.quantity * i.unit_price) AS revenue

FROM order_log o

JOIN order_details d ON o.order_id = d.order_id

JOIN item_catalog i ON d.item_id = i.item_id

GROUP BY month**;**

Output:

```
SQL> SELECT TO_CHAR(o.order_date, 'YYYY-MM') AS month,
  2         SUM(d.quantity * i.unit_price) AS revenue
  3  FROM order_log o
  4  JOIN order_details d ON o.order_id = d.order_id
  5  JOIN item_catalog i ON d.item_id = i.item_id
  6  GROUP BY TO_CHAR(o.order_date, 'YYYY-MM')
  7  ORDER BY month;

MONTH      REVENUE
------- ----------
2026-03       3992
2026-04       6690
```

4:Write a query to select Category-wise Sales

Query: SELECT i.item_group,

SUM(d.quantity * i.unit_price) AS total_sales

FROM order_details d

JOIN item_catalog i ON d.item_id = i.item_id

GROUP BY i.item_group;

Output:

```
SQL> SELECT i.item_group,
  2         SUM(d.quantity * i.unit_price) AS total_sales
  3  FROM order_details d
  4  JOIN item_catalog i ON d.item_id = i.item_id
  5  GROUP BY i.item_group;

ITEM_GROUP                                        TOTAL_SALES
------------------------------------------------- -----------
Stationery                                               1196
Accessories                                              3192
Electronics                                              6294
```

5:Write a query to select Inactive Customers

Query:

SELECT c.client_name

FROM client_info c

LEFT JOIN order_log o ON c.client_id = o.client_id

WHERE o.order_id IS NULL;

Output:

```
SQL> SELECT c.client_name
  2  FROM client_info c
  3  LEFT JOIN order_log o ON c.client_id = o.client_id
  4  WHERE o.order_id IS NULL;

CLIENT_NAME
-------------------------------------------------
Karan Mehta
Priya Nair
Amit Verma
Rahul Sharma
Sneha Reddy
```